

stocks recorded an average RoI of 78.79 percent during that period.

“There are two metrics that the investors or top management of a fab need to be worried about. One is what the load of the fab is and second is what is the yield on the fab. ‘Load’ is the function of demand or a company’s capability to sell the 30K to 40K wafer starts per month. ‘Yield’ is how good are the number of ‘usable’ chips from each 300mm wafer. The fabs which have made the chips of these high performing companies are very mature ones. They have already solved the yield problem on the well established and mature nodes, which is where the extraordinary growth is coming from. These fabs have also made significant investment into tooling for the newer nodes, which is where future demand will come. Note that in a good year a fab is a money printing machine,” says PVG Menon, former President of the India Electronics & Semiconductor Association (IESA), and former CEO, Electronic Sector Skills Council of India (ESSCI).

He adds, “Fabs announcing 100s of billions of dollars in R&D is proof in itself that fabs are money generating machines. Someone like a TSMC will not commit big amounts to R&D and capacity expansion unless it has demand (‘Load’) commitment from the likes of large-volume customers like Apple. Investing in a new node means investing at least \$4 billion per node.”

Shortage and high demand

Starting 2020, semiconductors have been in high demand and, despite many fabs being allowed to function amidst coronavirus lockdowns, their supply in the market continues to be scarce. This is primarily due to the constraints the semiconductor supply chain went through, and continues to go through, due to the lockdowns. After all, fabricating semiconductors is one thing and getting them delivered to organisations is another!

In plain economics, obviously, a market shortage occurs when there is excess demand. In such situations, customers are usually unable to buy as much as they require, neither are the manufacturers able to meet all the demands. Even though most of the semiconductor companies were not

Increase in Share Price Over a Year

Table 1

Semiconductor company	Price per share on 28th June 2020 (in \$)	Price per share on 28th June 2021 (in \$)
NVIDIA Corporation (NVDA)	366.2	761.24
Taiwan Semiconductor Manufacturing Corporation (TSMC)	56.33	116.53
Lattice Semiconductor	27.53	55.87
On Semiconductor	19.09	37.51
NXP Semiconductor	108.15	207.17

able to keep up with the demand of the semiconductors, their share prices continued to, and continue to, grow as of today.

“There is huge demand for semiconductors, and it is clear from every part of the supply chain that there are shortages. There is a very big gap between the demand and supply. It is quite possible that customers are paying higher prices to source semiconductors than what they have been usually paying in the normal times,” says Hemant Mallapur, co-Founder, Saankhya Labs.

He adds, “This works in favour of both fabless as well as fab semiconductor companies. Primarily, if you trace it back to the origin, you will find that during the first wave of Covid-19 there was a big slump in demand. Immediately the people noticed that such things can hammer their business. Also because of the backlog many companies ordered semiconductors in amounts more than what they would normally buy.”

The shares of Nvidia, a fabless semiconductor company, yielded 107.87 percent RoI during the one year period under review. By fabless, as most of you know, we mean a semiconductor company that owns the IP but does not own a semiconductor fab. Such companies outsource semiconductor manufacturing to vendors like Taiwan Semiconductor Manufacturing Corporation (TSMC).

Nvidia had a market cap of \$474.25 billion on 28th June 2021. Its stock price rose from \$366.2 per share (June 2020) to \$761.24 per share (June 2021).

Similarly, shares of Qualcomm, another fabless company, delivered approximately 56.38 percent RoI during the time frame. As a matter of fact, fabless semiconductor firms represented 32.8 percent of the total

market in 2020, a record for the segment, according to data from IC Insights.

Qualcomm had a market cap of \$155.27 billion on 28th June 2021. Its stock prices rose from \$88.02 per share (June 2020) to \$137.65 per share (June 2021).

The new record for market share for fabless companies was led by Qualcomm’s \$5 billion sales increase in 2020. IC Insights has also forecast that fabless companies and IC foundries will continue to be a strong force in the total IC industry landscape with a percentage share of the total IC market expected to reach the mid-30s in the next five years.

This shortage of semiconductors can primarily be linked to various factors including restraints in supply of raw materials used to manufacture these, restraint in supply chain that supplies semiconductors to customers, and high demand of these due to increased production of electronics.

“The demand for semiconductors is being driven by everything that has a connection with the end consumers. The increased demand of wearables, especially devices like pulse oxymeters, fitness bands, and smartwatches. The explosion in demand for computing devices like laptops, tablets, and smartphones due to massive increase in both work and study from home, is another factor. Then there is demand for relatively newer gadgets (in India) like dishwashers coming in. We are spending more and more on smart gadgets,” explains PVG Menon.

The 5G factor

The advancement of 5G is leading to a more connected world than ever. And enabling electronics with better connectivity features requires better and the latest technology based semiconductors. As the 5G technology con-